WORKED SOLUTION

# STEP III 1994, Question 4

Pure • Singular Solutions of ODEs

**STEP III 1994, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 8.1, *When Solutions Aren't Unique*: https://beyondstep.keithmcnulty.org/ch08/non-uniqueness.html

## The key idea

Take square roots: the equation says $\mathrm{d}y/\mathrm{d}x = \pm 2\sqrt{y}$, which separates at once. Better still, it is worth noticing that every parabola $y = (x-c)^2$ satisfies the equation, so the solutions are simply copies of $y = x^2$ slid along the $x$-axis. Once you see that picture, the later parts are about which copies pass through which points, and what all the copies have in common.

## Solution

### The general solution

Since $(\mathrm{d}y/\mathrm{d}x)^2 \geq 0$, any solution has $y \geq 0$. Where $y > 0$ we may write

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \pm 2\sqrt{y} \quad \Longrightarrow \quad \frac{1}{2\sqrt{y}}\frac{\mathrm{d}y}{\mathrm{d}x} = \pm 1 \quad \Longrightarrow \quad \sqrt{y} = \pm(x-c)$$

for a constant $c$, and squaring either sign gives

$$y = (x-c)^2.$$

It is worth checking directly that this whole parabola is a solution, including the left- and right-hand halves and the vertex: $y' = 2(x-c)$, so $(y')^2 = 4(x-c)^2 = 4y$ for every $x$.

**Ask yourself** The two signs gave the same formula. So where has the "$\pm$" gone?

It is hidden in the two halves of each parabola. On $y = (x-c)^2$ the gradient $2(x-c)$ is $-2\sqrt{y}$ to the left of the vertex and $+2\sqrt{y}$ to the right, so a single parabola uses both signs. The choice of sign reappears as a choice of $c$ when we fit a point.

### Solutions through $(a, b^2)$

We need $(a-c)^2 = b^2$, so $a - c = \pm b$, that is $c = a - b$ or $c = a + b$. The two solutions are

$$y = (x - a + b)^2 \qquad \text{and} \qquad y = (x - a - b)^2.$$

These are different because $b \neq 0$ makes $a - b \neq a + b$. (If $b = 0$ they would merge into one parabola with vertex at $(a, 0)$; the point would then also lie on the solution $y = 0$, which we meet again at the end.) Replacing $b$ by $-b$ just swaps the two, as it should, since the point $(a, b^2)$ is unchanged.

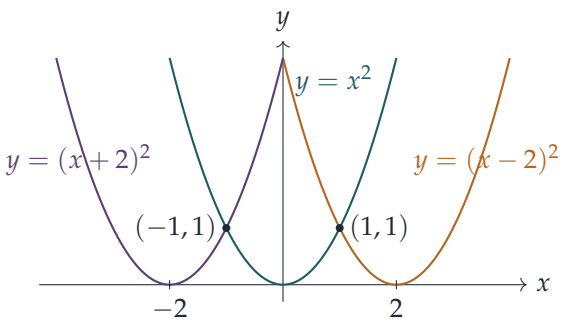
### The points $(a_1, 1)$ and $(a_2, 1)$

Take $b = 1$: through $(a, 1)$ the solutions are $y = (x - a + 1)^2$ and $y = (x - a - 1)^2$. A parabola $y = (x - c)^2$ passes through the origin exactly when $c = 0$, so we need $a - 1 = 0$ or $a + 1 = 0$. Hence

$$a_1 = 1, \qquad a_2 = -1.$$

Through $(1, 1)$ the solutions are $y = x^2$ (from $c = a_1 - 1 = 0$) and $y = (x-2)^2$. Through $(-1, 1)$ the solutions are $y = (x+2)^2$ and $y = x^2$ (from $c = a_2 + 1 = 0$).

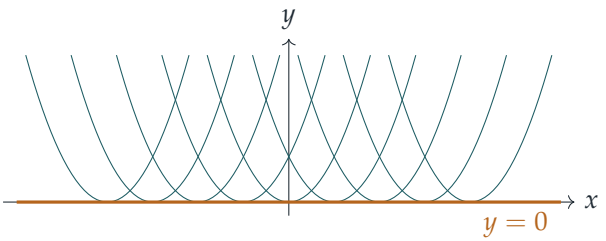
The solutions through the origin are, for both points, the same curve $y = x^2$, so their graphs coincide. This is no surprise: $(1, 1)$ and $(-1, 1)$ are the two points of $y = x^2$ at height 1, one on each side of the vertex. The other two solutions are $y = (x-2)^2$ and $y = (x+2)^2$, which are $y = x^2$ shifted 2 to the right and to the left.




At $(1, 1)$ the curve $y = x^2$ has gradient $+2$ and $y = (x-2)^2$ has gradient $-2$: the two solutions through a point always cross there with gradients $\pm 2b$, one for each square root.

**The family and its common tangent**

Varying $a$ and $b$ just varies $c = a \pm b$ over all real numbers, so the family is every horizontal translate of $y = x^2$.




Every parabola touches the $x$-axis at its vertex $(c, 0)$, where the gradient is 0. So the line $y = 0$ is a common tangent to the whole family (it is their *envelope*).

The relevance is that $y = 0$ is itself a solution, since then $y' = 0$ and $(y')^2 = 0 = 4y$. But it is not one of the parabolas for any $c$: it is a *singular solution*, which the separation of variables missed because we divided by $\sqrt{y}$. At each point $(c, 0)$ the solution $y = 0$ and the parabola $y = (x - c)^2$ have the same value and the same gradient, so solutions through points of the $x$-axis are not unique. In fact we can splice: for $p < q$,

$$y = \begin{cases} (x-p)^2, & x < p, \\ 0, & p \le x \le q, \\ (x-q)^2, & x > q \end{cases}$$

has a continuous derivative and satisfies the equation everywhere. So strictly the "two solutions" through $(a, b^2)$ are the two that agree with the parabolas near that point; away from it, once a solution reaches the $x$-axis, it may run along the axis for a while before leaving.

## Checks

- Substitute back: for $y = (x - a \pm b)^2$, at $x = a$ we get $y = b^2$, and $(y')^2 = 4y$ holds identically.
- The two gradients at $(a, b^2)$ are $2(a - c) = \pm 2b$, whose squares are both $4b^2 = 4y$, as required.
- Numerically, a central-difference derivative confirmed $(y')^2 = 4y$ for random parabolas and for the spliced curve above with $p = -1$, $q = 1$, and that the solutions through $(1, 1)$ and $(-1, 1)$ have $c \in \{0, 2\}$ and $c \in \{-2, 0\}$, sharing only $c = 0$.

## Common slips

- Writing $y' = 2\sqrt{y}$ only, and so getting just one half-parabola through each point.

- Forgetting $y = 0$: it is a genuine solution, and it is exactly the common tangent the last part is after.
- Thinking the two solutions through $(a_1, 1)$ and $(a_2, 1)$ that pass through the origin are different curves; both are $y = x^2$.

## Beyond STEP

This equation is the standard example of where uniqueness of solutions breaks down. The Picard–Lindelöf theorem guarantees exactly one solution of $y' = f(x, y)$ through a point when $f$ is Lipschitz in $y$, but $\pm 2\sqrt{y}$ is not Lipschitz near $y = 0$, and that is precisely where the parabolas and the singular solution meet. Envelopes of one-parameter families of curves, and the singular solutions they give for equations like Clairaut's, are a classical topic in first courses on ODEs. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness with examples like this one.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.